Functions and their graphs

Mathematics - Sample for API requests

What a function is

A function f: A -> B gives every element x of the set A exactly one element f(x) of the set B. A is the domain, B is the codomain, and the set of all values f(x) is the image of f.

The graph of f is the set of points (x, f(x)) in the plane. A vertical line meets the graph of a function at most once.

The linear function

f(x) = ax + b. Its graph is a straight line: two points are enough to draw it, usually where it meets the axes, (0, b) and (-b/a, 0).

- a > 0: the function is increasing.
- a < 0: the function is decreasing.
- a = 0: the function is constant and its graph is parallel to the x axis.

The quadratic function

f(x) = ax^2 + bx + c, with a not 0. Its graph is a parabola with the vertex V(-b/(2a), -D/(4a)). The parabola opens upwards when a > 0 (the vertex is a minimum) and downwards when a < 0 (the vertex is a maximum).

1. Find the vertex and the axis of symmetry x = -b/(2a).
2. Find where the graph meets the axes: (0, c), and the solutions of f(x) = 0, if any.
3. Make a table with two or three more values on each side of the vertex, then draw.

Exercises

1. Draw the graphs of f(x) = 2x - 4 and g(x) = -x + 1 and find where they meet.
2. Find the vertex of f(x) = x^2 - 6x + 5 and draw the parabola.
3. Find the maximum of f(x) = -x^2 + 4x + 1.